

Adding Story Problems

Answer Key

Kindergarten–Grade 1

Quick Answers

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|------------------------------------|------------------------------------|
| 1. how many crayons = 5 crayons, — | 5. how many buttons = 8 buttons, — |
| 2. 7 | 6. 10 pages |
| 3. 9 shells | 7. 10 |
| 4. 8 | 8. how many stamps = 9 stamps, — |
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What is verified

5 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 5, 8. The worked solution states what a correct response must contain.

The author marked 6 blanks as working space (*not* answers — nothing checks what lands there).

Curriculum

Level: Kindergarten–Grade 1

1.OA.C.6 – *problems 3, 6, 8*

K.OA.A – *problems 1, 2, 4, 5, 7*

Difficulty 1–4 of 5 across 11 tagged checks.

1. Leo has 3 crayons. Ana gives him 2 more crayons. Draw a circle for each crayon. How many crayons does Leo have now?

Solution: Act the story out: draw 3 circles for Leo’s crayons, then 2 more circles for the ones Ana gives him. Count them all: 1, 2, 3, 4, 5. So $3 + 2 = 5$. 5 crayons

Drawing (instructor judges): a good drawing shows 3 circles plus 2 more circles — 5 in all — matching the story.

2. There are 5 frogs on a log. 2 more frogs hop up onto it. Write the story numbers in the equation, then write how many frogs there are in all.

Solution: The story starts with 5 frogs and 2 more come, so the equation uses the story numbers 5 and 2: $5 + 2 = ?$ Count on from 5: 6, 7. So $5 + 2 = 7$. 7

3. Maya has 6 shells in her pail. She finds 3 more shells on the beach. Count on from 6 to find how many shells she has now.

Solution: Start at 6 and count on 3 more: 7, 8, 9. So $6 + 3 = 9$. 9 shells

4. There are 4 books in the bin. Dad puts in 4 more books. Write the story numbers in the equation, then write how many books are in the bin now.

Solution: Both story numbers are 4, so the equation is $4 + 4 = ?$ That is the double of 4: $4 + 4 = 8$. 8

5. The ten-frame shows Pia's 5 buttons. She gets 3 more buttons. Draw the 3 new buttons in the frame. How many buttons does Pia have in all?

Solution: The frame's top row holds the 5 buttons. Drawing the 3 new buttons fills three boxes of the bottom row. Count all the dots: $5 + 3 = 8$. 8 buttons

Drawing (instructor judges): the student draws 3 more dots in the frame, so it shows 8 dots in all.

6. Nia read 3 pages of her book in the morning and 7 pages at night. Count on from the bigger number to find how many pages she read in all.

Solution: Start big: begin at 7 and count on 3 more: 8, 9, 10. Addition works in any order, so $3 + 7 = 7 + 3 = 10$. 10 pages

7. Milo sees 8 ants on the sidewalk. 2 more ants join them. Write the story numbers in the equation, then write how many ants there are now.

Solution: The story numbers are 8 and 2: $8 + 2 = ?$ Count on from 8: 9, 10. So $8 + 2 = 10$. 10

8. Challenge! Tara has 2 stamps. She gets 4 stamps from Gram and 3 stamps from Papa. How many stamps does Tara have now? Tell how you found it.

Solution: Add all three story numbers: $2 + 4 + 3$. One easy way: add $4 + 3 = 7$ first, then count on from 7: 8, 9. So $2 + 4 + 3 = 9$. 9 stamps

Tell how you found it (instructor judges): any correct plan counts — for example grouping $4 + 3$ first and then adding 2, or starting at the biggest number 4 and counting on 3, then 2.